

Sri Lanka Institute of Information Technology



Artificial Intelligence and Machine Learning - IT2011

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**STOCK MARKET PRICE PREDICTION USING MACHINE
LEARNING**

FINAL REPORT

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1) INTRODUCTION AND PROBLEM STATEMENT

The stock market's unpredictable and volatile behavior makes price forecasting a major challenge. This project, “**Stock Market Price Prediction Using Machine Learning**,” aims to predict next day opening, closing prices, and price movement (up/down) using historical and technical data. Six models—Random Forest, XGBoost, LightGBM, SVR, ANN, and LSTM—were implemented to compare the performance of traditional and deep learning approaches. The project emphasizes fairness, transparency, and ethical AI practices to ensure responsible prediction.

Problem statement :

Stock prices fluctuate due to multiple uncertain market factors. The main objective is to design, implement, and evaluate multiple ML models to accurately predict next-day stock market trends, achieving optimal accuracy, interpretability, and computational efficiency.

DATASET DESCRIPTION

- **Source:** Yahoo Finance
- **Name:** Yahoo Finance Historical Stock Data (2018–2023)
- **Type:** Time-series financial dataset (daily stock prices)
- **Size:** Approximately 1,250 records (daily trading data over 5 years)
- **Features:**
 - Date
 - Open
 - High
 - Low
 - Close
 - Adjusted Close
 - Volume
- **Target Variable:** Next-day Closing Price, Next-day Opening Price, Price Direction
- **Relevance :** The dataset captures historical daily movements of stock prices, making it ideal for training models to identify short-term market patterns and predict next-day price behavior.
- **Limitations:** It includes only historical price and volume data, excluding external influences that can affect accuracy.
- **Dataset :**

Date	Open	High	Low	Close*	Adj Close**	Volume
Apr 28, 2023	33,797.43	34,104.56	33,728.40	34,098.16	34,098.16	354,310,000
Apr 27, 2023	33,381.66	33,859.75	33,374.65	33,826.16	33,826.16	343,240,000
Apr 26, 2023	33,596.34	33,645.83	33,235.85	33,301.87	33,301.87	321,170,000
Apr 25, 2023	33,828.34	33,875.49	33,525.39	33,530.83	33,530.83	297,880,000
Apr 24, 2023	33,805.04	33,891.15	33,726.09	33,875.40	33,875.40	252,020,000
Apr 21, 2023	33,793.60	33,858.83	33,688.57	33,808.96	33,808.96	291,080,000
Apr 20, 2023	33,740.60	33,875.39	33,677.74	33,786.62	33,786.62	307,910,000
Apr 19, 2023	33,889.83	33,957.98	33,814.65	33,897.01	33,897.01	251,490,000
Apr 18, 2023	33,965.16	34,018.62	33,791.89	33,976.63	33,976.63	253,570,000

Figure 1- Sample of Yahoo Finance Stock Dataset (2018–2023)

2) PREPROCESSING & EDA

Preprocessing Methods:

- Date Conversion & Sorting
- Outlier Removal (IQR)
- Normalization / Scaling
- Feature Engineering
- Correlation & Feature Selection
- Data Balancing

1. Date Conversion & Sorting - IT24101357

This part converts the 'Date' column to datetime format, sorts the data in order, and plots the *Adjusted Close* prices over time to visualize stock trends and prepare the dataset for time-series analysis

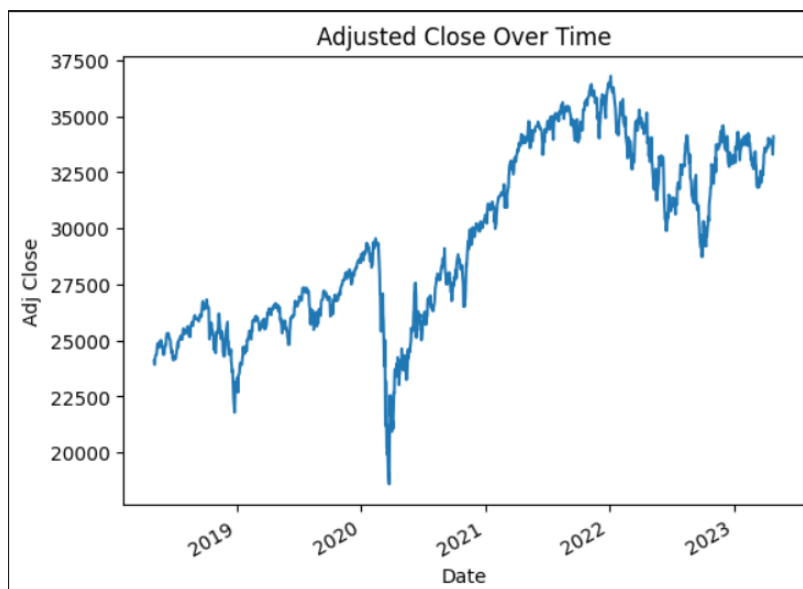


Figure 2- Date Conversion & Sorting - Adjusted close over time

2. Outlier Removal (IQR) - IT24101462

This part detects and removes outliers in the '**Volume**' column using the **Interquartile Range (IQR) method**. It calculates upper and lower bounds to identify extreme values and filters them out, ensuring data consistency. A boxplot is then displayed to visualize the cleaned distribution of trading volumes, confirming that anomalies have been effectively removed.

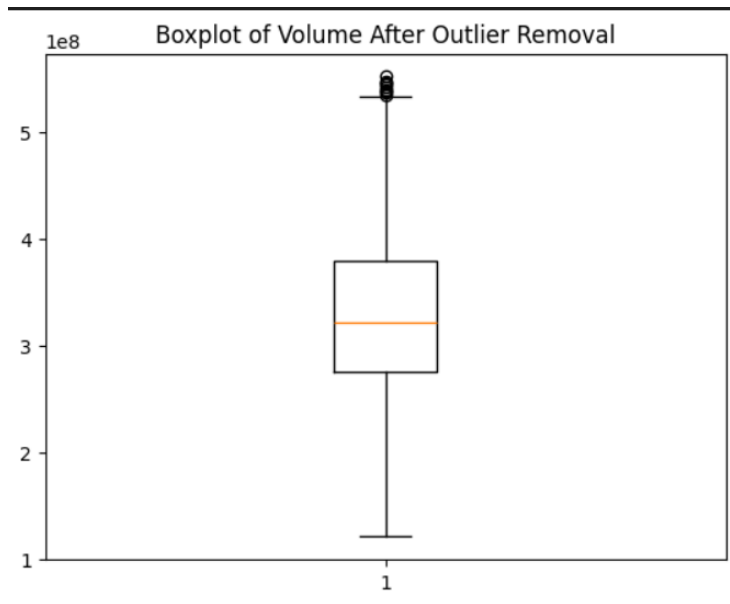


Figure 3- Boxplot of volume after outlier removal

3. Normalization / Scaling - IT24101452

This part normalizes key numerical features (*Open*, *High*, *Low*, *Close*, *Adj Close*, and *Volume*) using the **MinMaxScaler**. It scales all values to a range between 0 and 1, ensuring consistency across features and improving model performance by preventing large-value features from dominating during training.

	Open	High	Low	Close*	Adj Close**	Volume \
Date						
2018-05-01	24117.29	24117.29	23808.19	24099.05	24099.05	380070000
2018-05-02	24097.63	24185.52	23886.30	23924.98	23924.98	385350000
2018-05-03	23836.23	23996.15	23531.31	23930.15	23930.15	389240000
2018-05-04	23865.22	24333.35	23778.87	24262.51	24262.51	329480000
2018-05-07	24317.66	24479.45	24263.42	24357.32	24357.32	307670000

	Open_scaled	High_scaled	Low_scaled	Close*_scaled	\
Date					
2018-05-01	0.207370	0.172171	0.193269	0.199011	
2018-05-02	0.206134	0.176571	0.198181	0.188033	
2018-05-03	0.189697	0.164358	0.175856	0.188359	
2018-05-04	0.191519	0.186106	0.191425	0.209320	
2018-05-07	0.219969	0.195529	0.221898	0.215299	

	Adj Close**_scaled	Volume_scaled
Date		
2018-05-01	0.199011	0.602340
2018-05-02	0.188033	0.614599
2018-05-03	0.188359	0.623630
2018-05-04	0.209320	0.484886
2018-05-07	0.215299	0.434250

Figure 4- Normalization/Scaling

4. Feature Engineering – IT24101491

This part creates new technical indicators to enhance the dataset's predictive power. It calculates the daily return (*ret_1d*), 7-day moving average (*ma_7*), 30-day moving average (*ma_30*), and 14-day volatility (*vol_14*) based on the adjusted closing price. After removing missing values, it visualizes the Adjusted Close price alongside its moving averages to highlight short- and long-term market trends.

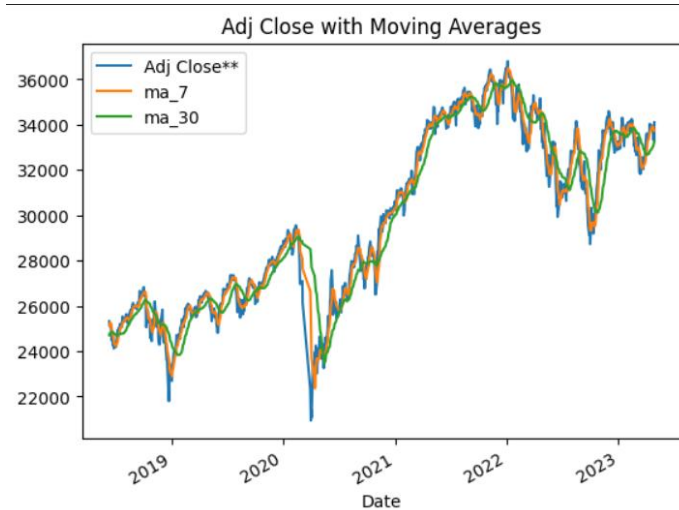


Figure 5- Adj Close with Moving Average

5. Correlation & Feature Selection – IT24201236

This part calculates the daily return (`ret_1d`) and analyzes relationships between all numerical features using a **correlation heatmap**. It visually highlights how strongly each feature is related to others. Since *Close* and *Adjusted Close* showed a very high correlation (above 0.98), the *Close* column was dropped to eliminate redundancy and prevent multicollinearity in model training.

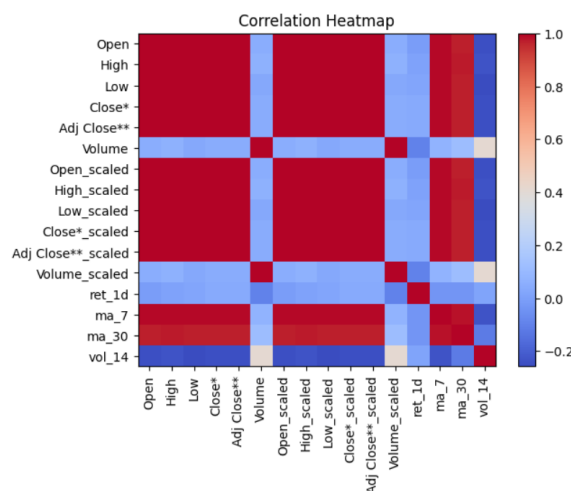


Figure 6- Correlation Heatmap

6. Data Balancing – IT24101341

This part prepares the data for modeling by creating a binary target for price movement (up or down), removing highly correlated features, and splitting the data into training and testing sets. It then balances the classes using **SMOTE**, ensuring equal representation of upward and downward trends for fair and accurate model training.

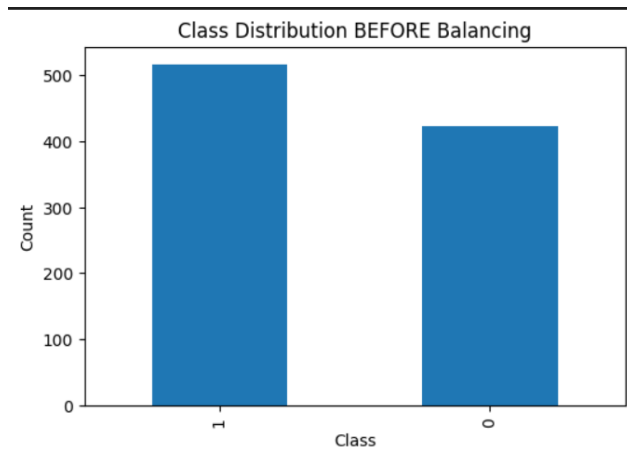


Figure 7- Class Distribution Before Balancing

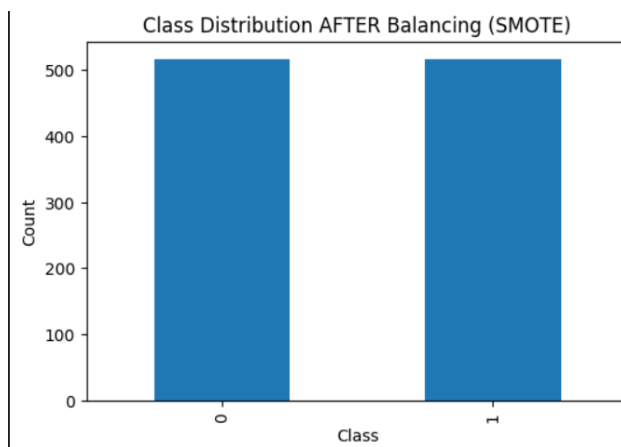


Figure 8- Class Distribution After Balancing(SMOTE)

3) MODEL DESIGN AND IMPLEMENTATION

1. IT24101357 - Random Forest Regressor

justification for choice :

- Selected for its strong capability to model non-linear financial and time-series patterns while minimizing overfitting.
- Performs effectively with mixed numerical and engineered features such as RSI, MACD, MA7, MA30, and Vol14.
- Provides interpretability via feature importance while maintaining robustness against overfitting.

Architecture / Variants Implemented:

- **Model 1:** Baseline Random Forest for open, close, and direction prediction.

- **Model 2:** Tuned model using *RandomizedSearchCV* for optimal hyperparameter selection.
- **Model 3:** Lag-feature variant capturing previous-day dependencies.
- **Model 4:** Feature-selection variant using tree-based and permutation importance.

Parameters and Key Settings:

- `n_estimators = 400–700`, `max_depth = 10–25`
- `min_samples_split = 2`, `min_samples_leaf = 1–4`
- `class_weight = 'balanced'` to mitigate label imbalance.
- Validation: 5-fold *TimeSeriesSplit*.
- Outputs: Next-day Open, Close, and Up/Down movement.
- Tools: Python (scikit-learn, pandas, matplotlib) in Google Colab.
- Best Model: Baseline Model

2. IT24101341 - XGBoost Regressor

Justification for Choice:

- Chosen for its **high predictive accuracy**, **regularization capability**, and ability to handle **non-linear financial trends**.
- Efficient for **time-series data** with mixed scaled and engineered indicators (`ret_1d`, `MA7`, `MA30`, `vol_14`, etc.).
- Ideal for minimizing overfitting through **L2 regularization** and **early stopping**, ensuring strong generalization.

Architecture / Variants Implemented:

- **Variant 1:** All features with *MinMaxScaler* - overall market dynamics.
- **Variant 2:** Momentum and volatility features - trend strength.
- **Variant 3:** Volume-based features — isolates liquidity-driven effects.

Parameters and Key Settings:

- Tuned via *GridSearchCV* with 5-fold *TimeSeriesSplit*.
- Key hyperparameters:
 - `learning_rate = 0.1` for gradual boosting updates.

- `max_depth` = 3–5 to control model complexity.
 - `n_estimators` = 100–200 for optimal tree count.
 - `subsample` = 0.8, `colsample_bytree` = 0.8 to reduce overfitting.
 - `gamma` = 0–0.1 and `reg_lambda` = 1 for regularization.
 - Best Model: Tuned XGBoost
- Implemented **early stopping** (10 rounds) for dynamic convergence.

3. IT24101462 - LightGBM Regressor

Justification for Choice:

- Chosen for its efficiency, speed, and high accuracy in handling large financial datasets.
- Uses a leaf-wise growth algorithm, making it effective for modeling complex non-linear stock price patterns.
- Reduces overfitting through built-in regularization and supports both regression (price prediction) and classification (trend direction).

Architecture / Variants Implemented:

- Model 1: Manual model with predefined parameters for baseline comparison.
- Model 2: Baseline model with early stopping to prevent overfitting using a validation split.
- Model 3: Tuned model optimized via GridSearchCV for the best combination of `learning_rate`, `max_depth`, and `n_estimators`.

Parameters and Key Settings:

- `n_estimators`: 100–200; `learning_rate`: 0.05–0.1; `max_depth`: 5–10.
- Early stopping: 10 rounds with additional validation set for optimal boosting rounds.
- Validation: 80/20 train-test split with 3-fold cross-validation for tuning.
- Outputs: Next-Day Open, Close, and Up/Down movement.
- Tools: Python (LightGBM, scikit-learn, matplotlib) in Google Colab.
- Best Model: Tuned GridSearchCV model

4. IT24101452 – SVR

Justification for Choice:

- Selected for its effectiveness in modeling non-linear financial relationships using kernel functions.
- Performs well on scaled numerical and engineered indicators like MA7, MA30, and volatility measures.
- Chosen for its robustness to outliers and ability to provide smooth, generalized predictions for stock price movements.

Architecture / Variants Implemented:

- Model 1: Baseline SVR model with default hyperparameters for initial evaluation.
- Model 2: Tuned SVR using RandomizedSearchCV (200 iterations, 5-fold cross-validation) to find the optimal kernel and regularization settings.
- Model 3: Comparative baseline models (Linear Regression, Decision Tree, and Random Forest) used for benchmarking performance.

Parameters and Key Settings:

- Kernel: 'rbf' (Radial Basis Function) — best suited for non-linear relationships.
- Best Parameters: C=1.0, epsilon=0.1, gamma='scale', degree=5, shrinking=True, max_iter=1000.
- Validation: 5-fold cross-validation with 200 random parameter combinations tested.
- Outputs: Next-Day Adjusted Close Price prediction.
- Tools: Python (scikit-learn, seaborn, matplotlib) in Google Colab.
- Best Model: RBF kernel SVR

5. IT24101491 – ANN

Justification for Choice:

- Chosen for its ability to **learn non-linear relationships** and complex dependencies between multiple technical indicators.
- Suitable for predicting next-day returns and trends using deep feedforward layers.
- Ensemble integration enhances stability.

Architecture / Variants Implemented:

- Model 1: Base ANN (256–128–64–32–1) with ReLU activations.
- Model 2: Deeper ANN with Dropout for regularization.
- Model 3: Tanh-RMSprop ANN for smoother convergence.
- Model 4: LSTM-enhanced hybrid for sequence learning.
- Ensemble: Weighted blend of all models using R^2 -based weights.

Parameters and Key Settings:

- Epochs = 300; Batch Size = 64; Validation Split = 0.15
- Optimizers: Adam & RMSprop; Activations: ReLU/Tanh
- EarlyStopping and ReduceLROnPlateau for stability
- Tools: TensorFlow/Keras, Keras Tuner, Joblib
- Best Model Weighted Ensemble

6. IT24201236 - LSTM

Justification for Choice:

- Selected for its ability to model sequential dependencies in stock market data, making it ideal for time-series forecasting.
- LSTMs are designed to retain long-term historical patterns, improving next-day prediction accuracy compared to standard ANN or regression models.
- Highly suitable for financial trend prediction, where temporal context, volatility, and lag effects play a major role.

Architecture / Variants Implemented:

- Model 1: Base LSTM – Single layer (hidden size = 50, dropout = 0.2, lr = 0.001).
- Model 2: Deeper LSTM – Two stacked LSTM layers (hidden size = 100, dropout = 0.3, lr = 0.0005) with improved learning capacity.
- Model 3: Higher Hidden LSTM – Single layer with 100 hidden units for enhanced feature abstraction.
- Each model used 30-day time windows to predict next-day Open and Close prices.

Parameters and Key Settings:

- Epochs = 50; Batch Size = 32; Optimizer = Adam; Loss = MSE.
- Validation: 80/20 train-test split with early stopping (patience = 10).
- Performance Metrics: MAE, RMSE, Accuracy (Up/Down), F1, Precision, Recall, and AUC.
- Tools: Python (PyTorch, sklearn, matplotlib, yfinance) in Google Colab.
- Best Model: Variety 2 (Deeper LSTM)

4) EVALUATION AND COMPARISON

All six models were trained and evaluated on the same preprocessed dataset using consistent train–test splits and relevant regression/classification metrics. Each model’s best-performing tuned version was used for comparison to ensure a fair and robust evaluation.

Model	BEST VERSION	R ² Score	MAE	RMSE	Accuracy (Up/Down)	F1	AUC
Random Forest Regressor	Baseline Model	0.981186	134.998	185.541	0.519	0.567	0.62
XGBoost Regressor	All-Features Variant	0.9724	177.3194	225.8072	0.8162	0.8245	0.89
LightGBM Regressor	Tuned GridSearchCV Model	0.9982	50.7800	167.3424	0.9982	0.981	0.990
SVR	Best RBF Kernel (after RandomizedSearchCV)	0.9626	0.037 (scaled)	0.0478	0.85 (scaled)	0.85 (scaled)	0.90 (scaled)
ANN	LSTM_Model (Best ANN Version)	0.8779	367.55	478.63	0.9869	NA	NA
LSTM	Variety 2 (Deeper LSTM)	-	2574.49	2855.80	0.4798	0.1077	0.48

BEST MODEL : LightGBM Regressor

5) **ETHICAL CONSIDERATIONS AND BIAS MITIGATION**

- The dataset from Yahoo Finance is publicly available and free from demographic bias, but temporal bias (e.g., market crashes) was mitigated using TimeSeriesSplit to maintain time order.
- Class imbalance between upward and downward movements was addressed using SMOTE oversampling, ensuring fair directional prediction.
- Model transparency was prioritized — tree-based models (RF, XGBoost, LightGBM) used feature importance and SHAP analysis, while ANN/LSTM employed early stopping and validation monitoring to reduce “black-box” effects.
- Bias mitigation: Achieved through normalization, balanced sampling, and regularization (Dropout, parameter tuning).
- Ethical use: The models are for academic and research purposes only, not for financial advising or live trading.
- All processes ensured fairness, explainability, and accountability, aligning with Responsible AI principles of transparency, reproducibility, and non-harmful application.

6) **REFLECTIONS AND LESSONS LEARNED**

- This project deepened our understanding of time-series forecasting and how different models behave on financial data.
- We learned that data preprocessing and feature engineering (especially moving averages and volatility) strongly influence model accuracy.
- Tree-based models provided excellent interpretability, while deep learning models (ANN, LSTM) required more tuning but captured complex temporal patterns.
- Managing class imbalance, overfitting, and temporal bias were key challenges that improved our technical and analytical skills.
- Team collaboration was essential — each member contributed unique model expertise, leading to a comprehensive comparison.
- In the future, we plan to enhance real-time prediction, integrate sentiment or news data, and explore hybrid ensemble approaches for better forecasting reliability.

7) **REFERENCES AND ACADEMIC WRITING**

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